

Specifying Bayesian Phylogenetic Models using LPhy

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Why LPhy ?

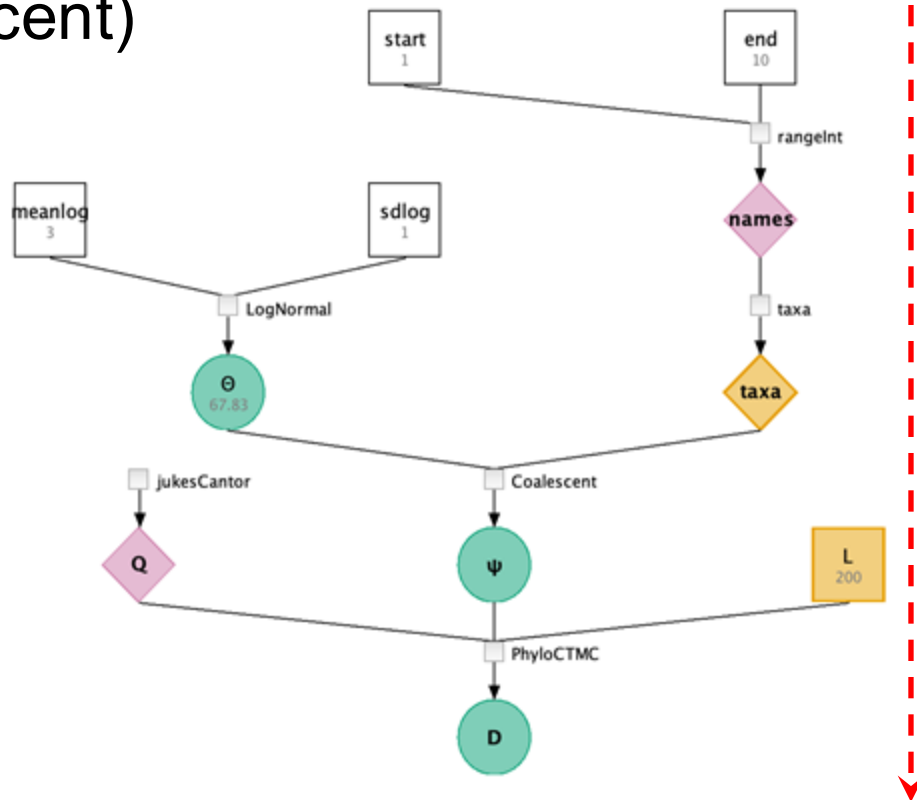
- Phylogenetic model specification.
- Designed to enhance the readability, reproducibility, and reusability of phylogenetic models.
- Probabilistic graphical models (PGM).
- LPhy studio – Graphical user interface + natural language narratives describing models + citations + a hub for connecting to other tools.
- Data simulations.
- LPhyBEAST – integration with the BEAST 2.
- Learning Bayesian phylogenetics.

LPhy & Studio (JC + Coalescent)

```
data {  
  L = 200;  
  taxa = taxa(names=1:10);  
}  
model {  
   $\theta \sim \text{LogNormal}(\text{meanlog}=3.0, \text{sdlog}=1.0);$   
   $\psi \sim \text{Coalescent}(\text{taxa}=\text{taxa}, \text{theta}=\theta);$   
   $D \sim \text{PhyloCTMC}(L=L, Q=\text{jukesCantor}(), \text{tree}=\psi);$   
}
```

Posterior

$$P(\psi, \theta | D) \propto P(D | \psi) P(\psi | \theta) P(\theta)$$



LPhy Language & Probabilistic Graphical Model

- Generator:

- Generative distribution:

```
~ LogNormal(meanlog=3.0, sdlog=1.0);
```

- Deterministic function:

```
=jukesCantor()
```

- Value:

- Constant:



- Variable:



Random variable:



Two Code Blocks : Data & Model

- Data block

- Observed data like the number of taxa, sequence length, observed traits, ...
- Defines all the data that will be supplied to the model from the outside (e.g., from a file, or preprocessed dataset).

- Model block

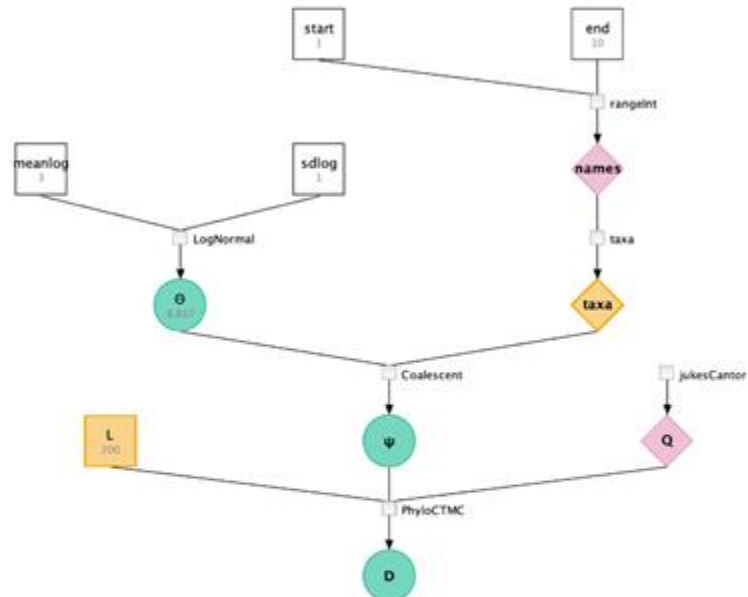
- Priors,
- Likelihood,
- Model logic

```
data {  
  L = 200;  
  taxa = taxa(names=1:10);  
}  
model {  
   $\theta$  ~ LogNormal(meanlog=3.0, sdlog=1.0);  
   $\psi$  ~ Coalescent(taxa=taxa, theta= $\theta$ );  
  D ~ PhyloCTMC(L=L, Q=jukesCantor(), tree= $\psi$ );  
}
```



AutoLayout Interactive

Model Narrative Latex Variable Summary Variable Log Tree Log Alignment Errors



jcCoalescent.lphy

Code

```
data {
  L = 200;
  taxa = taxa(names=1:10);
}
model {
  theta ~ LogNormal(meanlog=3.0, sdlog=1.0);
  psi ~ Coalescent(taxa=taxa, theta=theta);
  D ~ PhyloCTMC(L=L, Q=jukesCantor(), tree=psi);
}
```

Model

The alignment, D is assumed to have evolved under a phylogenetic continuous time Markov process (Felsenstein, 1981) on phylogenetic time tree, ψ , with an instantaneous rate matrix and alignment length, L of 200. The instantaneous rate matrix is the Jukes-Cantor model (Jukes and Cantor, 1969). The phylogenetic time tree, ψ is assumed to come from a Kingman's coalescent tree prior (Rodrigo and Felsenstein, 1999) with coalescent parameter, θ and $taxa$. The coalescent parameter, θ has a log-normal prior with a mean in log space of 3.0 and a standard deviation in log space of 1.0.

Posterior

$$P(\psi, \theta | D) \propto P(D | \psi) P(\psi | \theta) P(\theta)$$

References

- Felsenstein, J. (1981). Evolutionary trees from DNA sequences: a maximum likelihood approach. *Journal of molecular evolution*, 17(6), 368-376. <https://doi.org/10.1007/BF01734359>
- Jukes, T. H., & Cantor, C. R. (1969). Evolution of protein molecules. *Mammalian protein metabolism*, 3, 21-132. <https://doi.org/10.1016/B978-1-4832-3211-9.50009-7>
- Rodrigo AG, Felsenstein J. (1999). Coalescent Approaches to HIV Population Genetics. *The Evolution of HIV*.

Include
Code
Model
Posterior
References

Exclude
Graphical Model
Data

reps: 1

Sample

Layering:

From Sinks



Show constants

100%

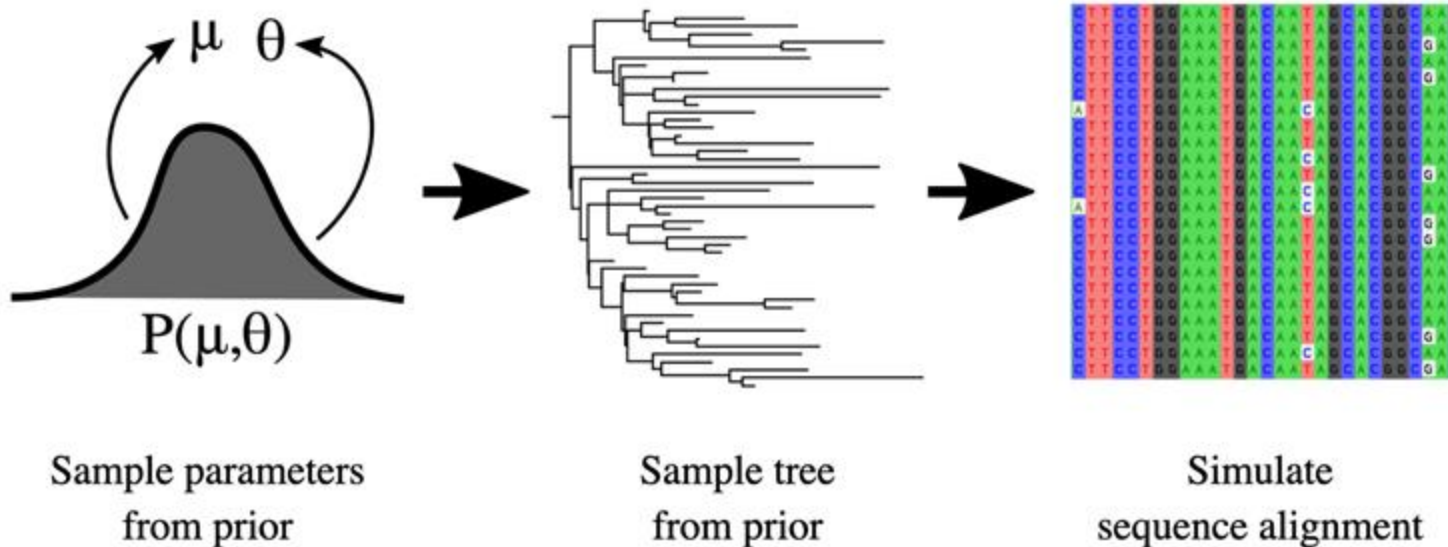
data model

```
1 theta ~ LogNormal(meanlog=3.0, sdlog=1.0);
2 psi ~ Coalescent(taxa=taxa, theta=theta);
3 D ~ PhyloCTMC(L=L, Q=jukesCantor(), tree=psi);
```

>

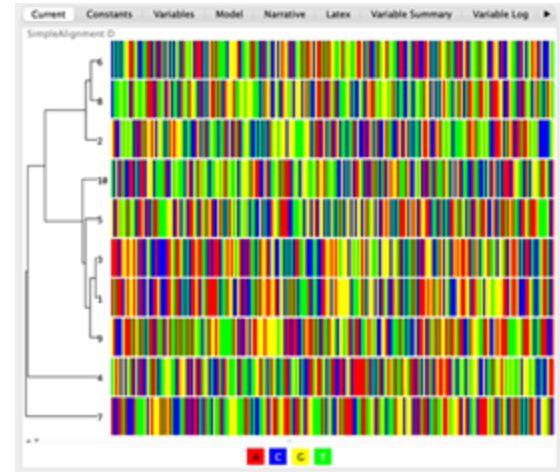
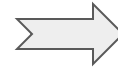
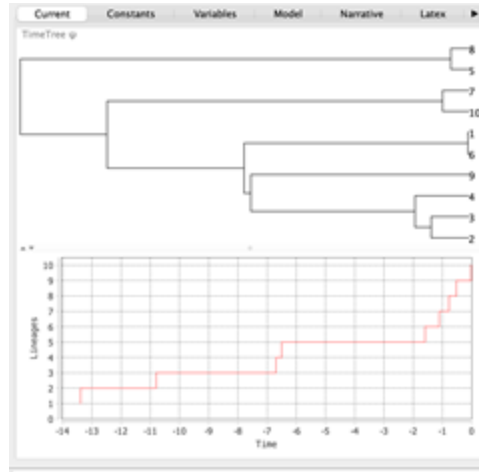
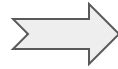
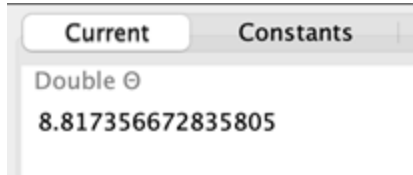
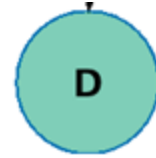
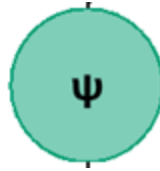
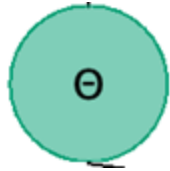
The Neutrality Assumption

Because of the way we've factorized the joint probability for the data and model parameters, we are implicitly assuming that our alignment could have been produced in the following fashion:



Simulation in LPhy

It's easier to understand using the Probabilistic Graphical Model (PGM). To simplify the view, uncheck "Show constants" in the Studio.



Multiple Simulations



	Latex	Variable Summary	Variable Log
Sample	Theta		
0	57.96138060222518		
1	2.9460709679484665		
2	5.019479911015483		
3	52.543037714410374		
4	30.219474105979977		
5	45.3208989593306		
6	43.72966491652776		
7	19.31147839098886		
8	9.494089811074023		
9	44.031335415561045		

	Latex	Variable Summary	Variable Log	Tree Log	Alignment	Errors
sample	psi					
0	((4:10.88871391351639,((1:1.9132619501079977,8:1.9132619501079977):1.135					
1	(((((10:0.16988113743499061,4:0.16988113743499061):0.05891200486985465,2					
2	((((5:0.10282768385992361,2:0.10282768385992361):1.9086397898039944,((10:0					
3	((((3:4.1090247827917965,6:4.1090247827917965):52.852004896370055,(8:14.0					
4	((2:7.023638057400394,(6:0.299232255344443,4:0.299232255344443):6.724405					
5	((10:17.08117250613943,5:17.08117250613943):122.59497177945502,((1:12.05					
6	((5:14.363139803927837,((1:2.490941162182306,7:2.490941162182306):4.9015					
7	(((((3:0.9483623420822407,9:0.9483623420822407):2.932946186691972,7:3.881					
8	((2:1.4656360663406032,(9:0.38773142694960666,5:0.38773142694960666):1.0					
9	((9:24.528330758374736,7:24.528330758374736):35.53026867656844,((3:9.484					

	Latex	Variable Summary	Variable Log	Tree Log	Alignment
sample	D				
0	nucleotide alignment 10 by 200				
1	nucleotide alignment 10 by 200				
2	nucleotide alignment 10 by 200				
3	nucleotide alignment 10 by 200				
4	nucleotide alignment 10 by 200				
5	nucleotide alignment 10 by 200				
6	nucleotide alignment 10 by 200				
7	nucleotide alignment 10 by 200				
8	nucleotide alignment 10 by 200				
9	nucleotide alignment 10 by 200				

Phylogenetic Models

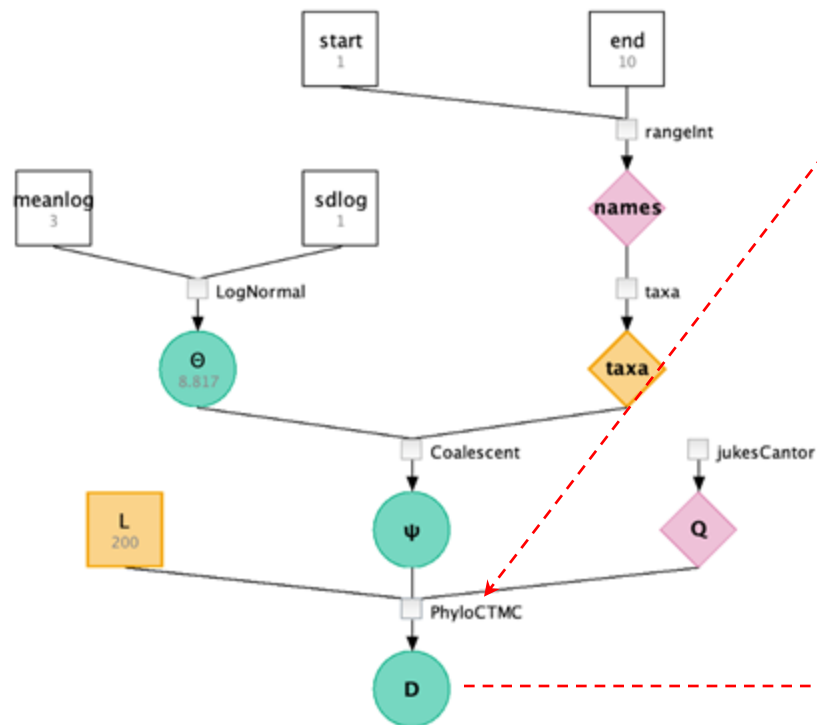
The Phylogenetic Likelihood

$$\Pr(D|T, \mu)$$

- Computed by Felsenstein's pruning algorithm
- D is a multiple sequence alignment of n sequences,
- T is a phylogenetic tree with n leaves, and
- μ are the parameters of the chosen CTMC-based model of sequence evolution.
- Just a product of site pattern probabilities: note the obvious generalizations to multiple alignments that share a tree, or to phylogenetic networks that allow different sites to evolve under distinct "local" trees.

We're Bayesians: we need a probability distribution for T !

AutoLayout Interactive



Current

Constants

Variables

Model

Narrative

Latex

PhyloCTMC(L=L, Q=jukesCantor(), tree=ψ);

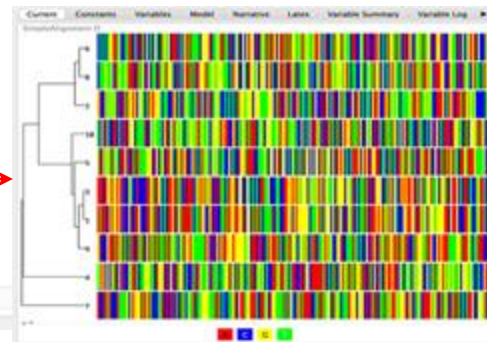
PhyloCTMC distribution

The phylogenetic continuous-time Markov chain distribution. A sequence is simulated for every parameters:

- tree ($\psi = ((8[p=0.0, p_{\max}=0.0, p_{\min}=0.0]:0.5212503227366986, 5[p=0.1111111111])$
- mu (null); the clock rate. Default value is 1.0.
- freq (null); the root probabilities. Optional parameter. If not specified then first row of e^Q
- Q $([[-1.0, 0.3333333333333333, 0.3333333333333333, 0.3333333333333333], [0.3333333333333333, -1.0, 0.3333333333333333, 0.3333333333333333], [0.3333333333333333, 0.3333333333333333, -1.0, 0.3333333333333333], [0.3333333333333333, 0.3333333333333333, 0.3333333333333333, -1.0]])$
- siteRates (null); a rate for each site in the alignment. Site rates are assumed to be 1.0 otherwise
- branchRates (null); a rate for each branch in the tree. Original branch rates are used if rat
- L (L = 200); length of the alignment
- dataType (null); the data type used for simulations, default to nucleotide
- root (null); root sequence, defaults to root sequence generated from equilibrium frequen

Reference

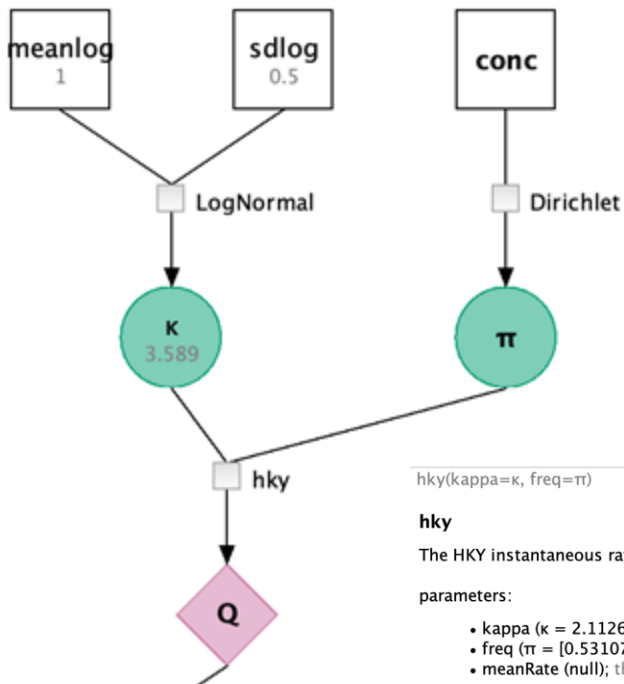
Felsenstein, J. (1981). Evolutionary trees from DNA sequences: a maximum likelihood approach. <https://doi.org/10.1007/BF01734359>



reps: 1 Sample Layering: From Sinks ☒ Show constants 100%

Substitution Model

$$Q = \text{hky}(\text{kappa} = \kappa, \text{freq} = \pi)$$



A
C
G
T

Current	Constants	Variable	
Double[][] Q			
$\begin{bmatrix} -1.31127 & 0.06958 & 1.11208 & 0.1296 \\ 0.19778 & -0.97275 & 0.30989 & 0.46508 \\ 0.70975 & 0.06958 & -0.90893 & 0.1296 \\ 0.19778 & 0.24971 & 0.30989 & -0.75738 \end{bmatrix}$			
A	C	G	T

hky(kappa=κ, freq=π)

hky

The HKY instantaneous rate matrix. Takes a kappa and base frequencies (and option

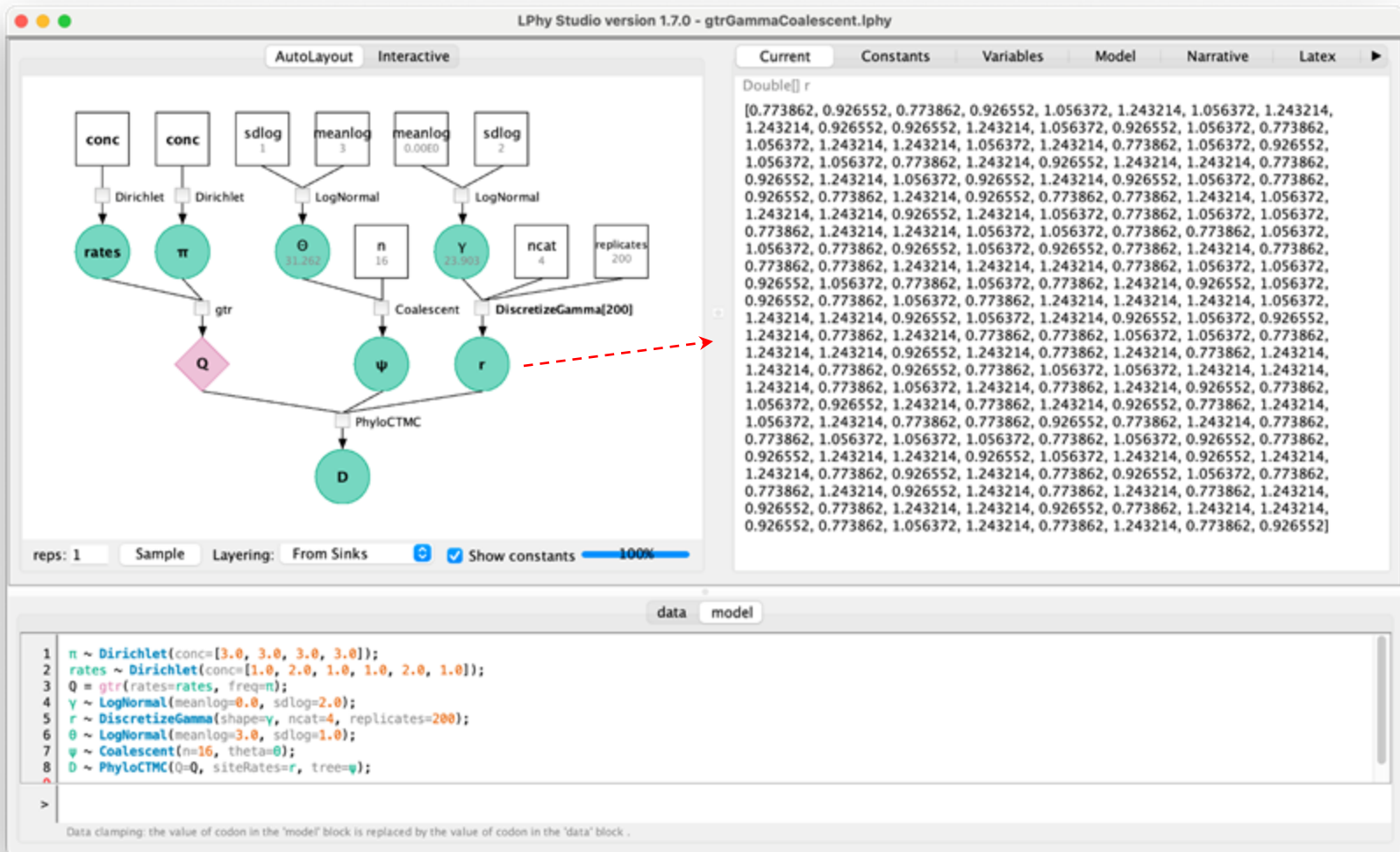
parameters:

- kappa (κ = 2.1126082492322333); the kappa of the HKY process.
- freq (π = [0.5310757421038967, 0.35389030790459847, 0.0613814047301, 0.05363395386536182]); the base frequencies.
- meanRate (null); the total rate of substitution per unit time. Default 1.0.

Reference

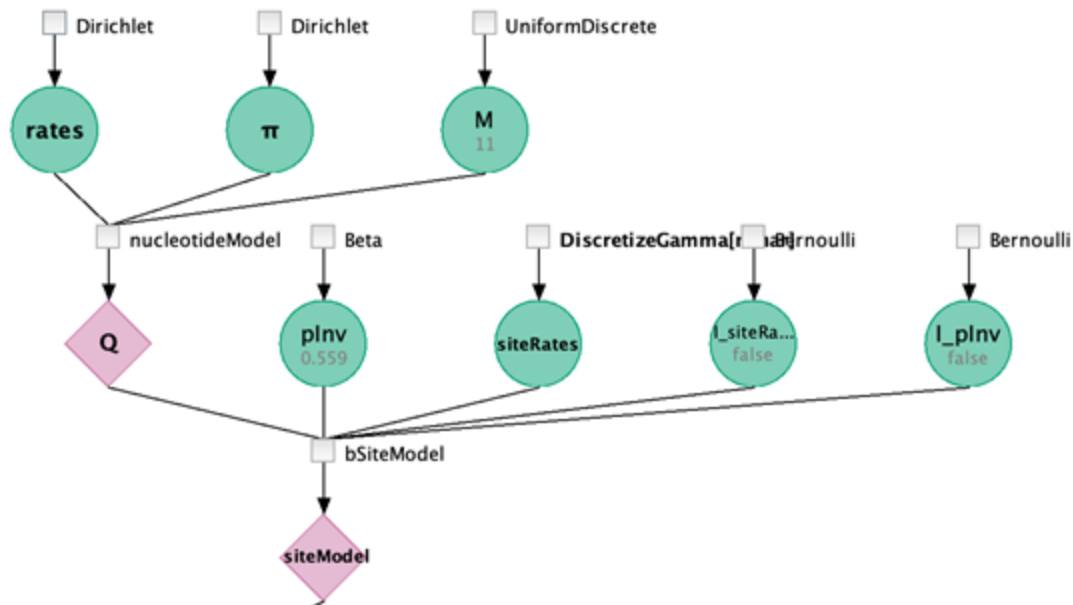
Hasegawa, M., Kishino, H. & Yano, T. (1985) Dating of the human-ape splitting by a
<https://doi.org/10.1007/BF02101694>

Site Rates



Site Model

Simple bModel Test:



```

 $\pi$  ~ Dirichlet(conc=[3.0, 3.0, 3.0, 3.0]);
modelSet = bModelSet("transitionTransversionSplit");
M ~ UniformDiscrete(lower=0, upper=modelSet.size()-1);
rates ~ Dirichlet(conc=[1.0, 2.0, 1.0, 1.0, 2.0, 1.0]);
Q = nucleotideModel(modelSet=modelSet, modelIndicator=M, rates=rates, freq= $\pi$ );
pInv ~ Beta(alpha=1.0, beta=1.0);
siteRates ~ DiscretizeGamma(shape=0.5, ncat=4, replicates=nchar);
I_pInv ~ Bernoulli(p=0.5);
I_siteRates ~ Bernoulli(p=0.5);
siteModel = bSiteModel(Q=Q, siteRates=siteRates, pInv=pInv, useSiteRates=I_siteRates, usePInv=I_pInv);

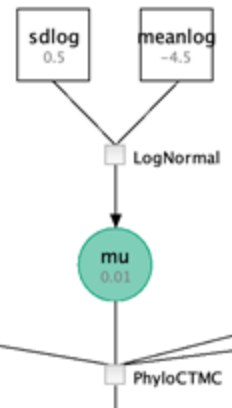
```

Molecular Clock Model

- Strict clock : **mu**, default to 1

```
mu ~ LogNormal(meanlog=-4.5, sdlog=0.5);
```

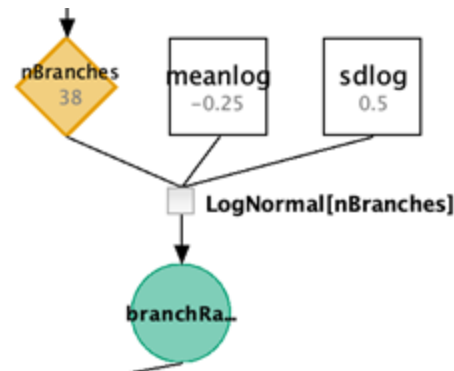
```
D ~ PhyloCTMC(L=200, Q=jukesCantor(), mu=mu, tree=ψ);
```



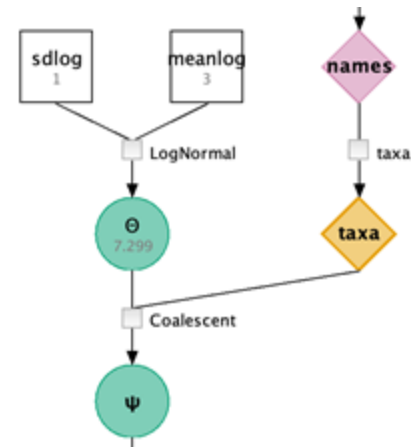
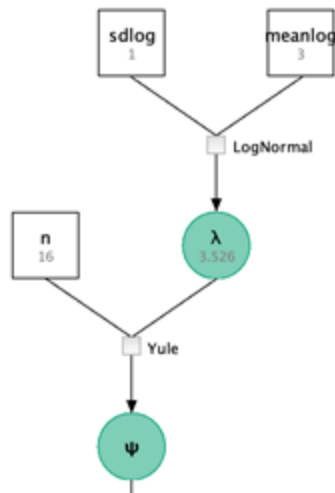
- Simple relaxed clock : **branchRates**, a rate for each branch

```
branchRates ~ LogNormal(meanlog=-0.25, sdlog=0.5, replicates=nBranches);
```

```
D ~ PhyloCTMC(L=100, Q=jukesCantor(), branchRates=branchRates, tree=ψ);
```

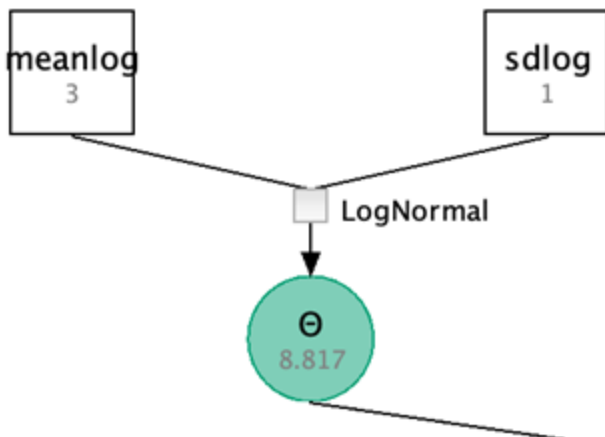


Tree Prior



Feature	Birth–Death Tree	Coalescent Tree
Parameters	Speciation rate (λ), Extinction rate (μ)	Effective population size (N_e)
Time Direction	Forward (speciation)	Backward (coalescence)
Branch Lengths	Longer internal branches	Shorter internal branches (especially recent)
Branching Pattern	More balanced (depends on λ/μ)	Often unbalanced (more recent coalescence)
Used for	Species-level trees	Gene trees and population-level trees

Parameter Priors



- Prior is the part of your model;
- Why use a **LogNormal** distribution instead of a **Normal** distribution for rates?
- Why do we try to avoid using a **Uniform** prior?

➤ Tricks in R:

```
qlnorm(p=c(0.025, 0.5, 0.975), meanlog=3, sdlog=1)
```

```
2.829319 20.085537 142.588660
```

Multiple Partitions

Unlinking models

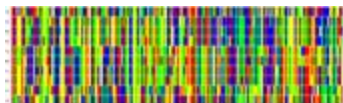


HKY

Strict Clock

Tree

```
Coalescent(..., theta=⊗);
```



HKY

Strict Clock

Tree

```
Coalescent(..., theta=⊗);
```



HKY

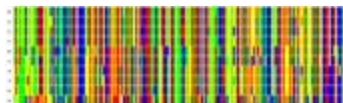
Strict Clock

Tree

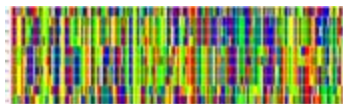
```
Coalescent(..., theta=⊗);
```

Multiple Partitions

Linking models is essentially about sharing parameters across models.



HKY



HKY

Stric Clock

Tree

```
Coalescent(..., theta=⊕);
```



HKY

Relative rate

Relative rate

×

Clock rate

Relative rate

Vectorization

1. Use an optional 'replicates' argument to generate vectors of independent and identically distributed (IID).
2. Implicit vectorization similar to R, allowing all arguments of distributions and functions to be vectorised.

```
 $\pi$  ~ Dirichlet(conc=[2.0, 2.0, 2.0, 2.0], replicates=3);
```

```
 $\kappa$  ~ LogNormal(meanlog=1.0, sdlog=0.5, replicates=3);
```

```
L = [209, 210, 210];
```

```
 $r$  ~ WeightedDirichlet(conc=rep(element=1.0, times=3), weights=L);
```

```
Q=hky(kappa= $\kappa$ , freq= $\pi$ , meanRate= $r$ )
```

Sequence Types

- Nucleotide
- Amino acid
- Binary
- Standard sequence type
 - Define the number data states

`dataType=standard(numStates=4),`

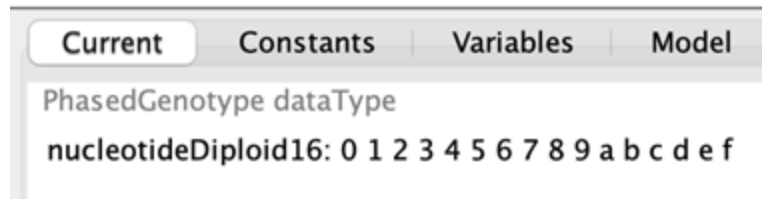
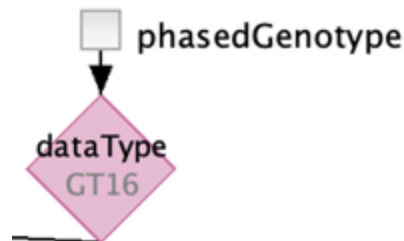
- Import, e.g., H5N1.lphy

`dataType = D_trait.dataType()`

LPhy extension “Phylonco”

- GT16 sequence type

`dataType=phasedGenotype()`



AutoLayout Interactive

Current Constants Variables Model Narrative ▶

standard(numStates=4)

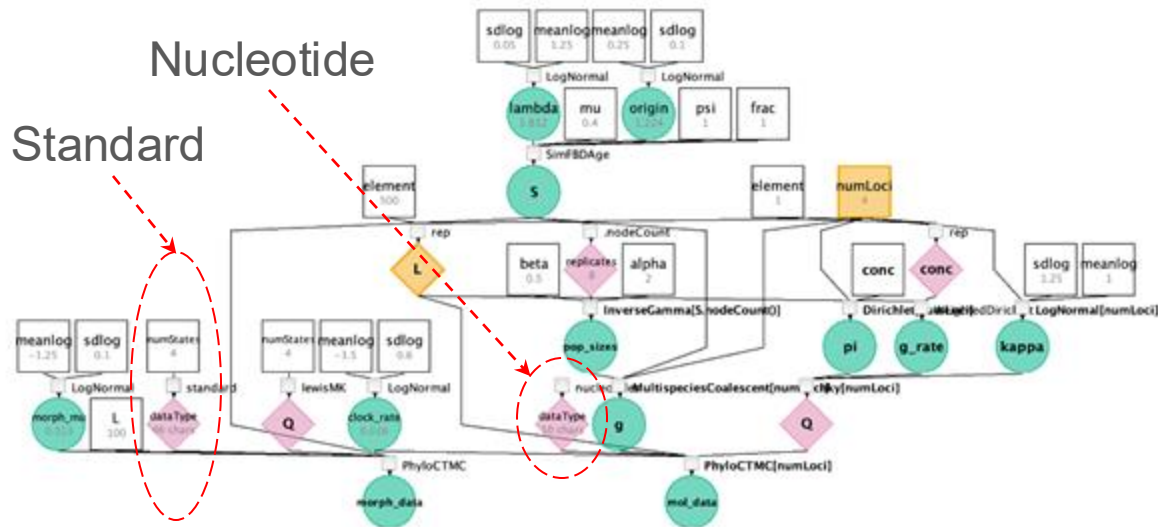
standard

The Standard data type function. Takes a state count and produces a Sta

parameters:

- numStates (4); the number of distinct states in this data type.

Nucleotide
Standard


 reps: 1 Sample Layering: From Sinks ☒ Show constants 100%

data model

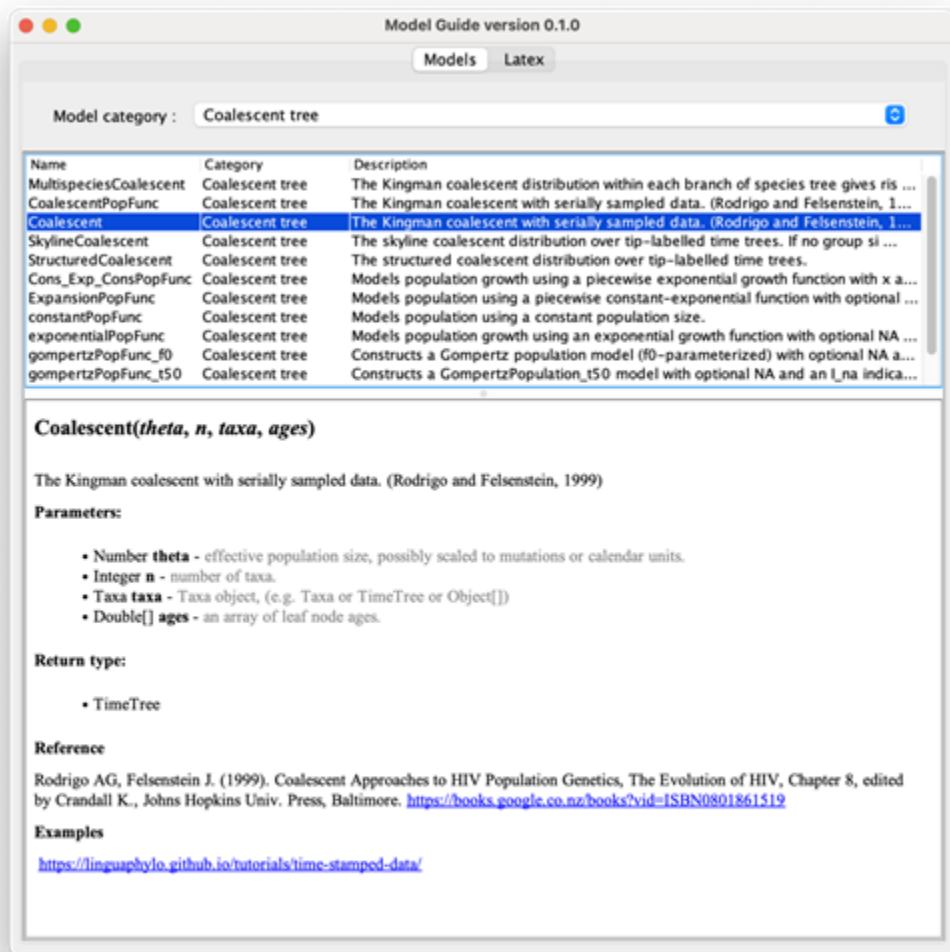
```

1 pi ~ Dirichlet(conc=[5, 5, 5, 5], replicates=numLoci);
2 kappa ~ LogNormal(meanlog=1.0, sdlog=1.25, replicates=numLoci);
3 g_rate ~ WeightedDirichlet(conc=rep(element=1.0, times=numLoci), weights=L);
4 clock_rate ~ LogNormal(meanlog=-3.5, sdlog=0.6);
5 lambda ~ LogNormal(meanlog=1.25, sdlog=0.85);
6 origin ~ LogNormal(meanlog=0.25, sdlog=0.1);
7 S ~ SimfBDAge(frac=1.0, lambda=lambda, mu=0.4, originAge=origin, psi=1.0);
8 pop_sizes ~ InverseGamma(alpha=2.0, beta=0.5, replicates=S.nodeCount());
9 g ~ MultispeciesCoalescent(S=S, theta=pop_sizes, replicates=numLoci);
10 mol_data ~ PhyloCTMC(L=L, Q=Qxy(kappa=kappa, freq=pi, meanRate=g_rate), dataType=nucleotides(), mu=clock_rate, tree=g);
11 morph_mu ~ LogNormal(meanlog=-1.25, sdlog=0.1);
12 morph_data ~ PhyloCTMC(L=100, Q=lewis98(numStates=4), dataType=standard(numStates=4), mu=morph_mu, tree=S);

```

Doc

- <https://linguaphylo.github.io>
[/linguaPhylo/](https://linguaphylo.github.io)
- **Model Guide** from Tools menu



Model Guide version 0.1.0

Models Latex

Model category : Coalescent tree

Name	Category	Description
MultispeciesCoalescent	Coalescent tree	The Kingman coalescent distribution within each branch of species tree gives ris ...
CoalescentPopFunc	Coalescent tree	The Kingman coalescent with serially sampled data. (Rodrigo and Felsenstein, 1...
Coalescent	Coalescent tree	The Kingman coalescent with serially sampled data. (Rodrigo and Felsenstein, 1...
SkylineCoalescent	Coalescent tree	The skyline coalescent distribution over tip-labelled time trees. If no group si ...
StructuredCoalescent	Coalescent tree	The structured coalescent distribution over tip-labelled time trees.
Cons_Exp_ConsPopFunc	Coalescent tree	Models population growth using a piecewise exponential growth function with x a...
ExpansionPopFunc	Coalescent tree	Models population using a piecewise constant-exponential function with optional ...
constantPopFunc	Coalescent tree	Models population using a constant population size.
exponentialPopFunc	Coalescent tree	Models population growth using an exponential growth function with optional NA ...
gompertzPopFunc_f0	Coalescent tree	Constructs a Gompertz population model (f0-parameterized) with optional NA a...
gompertzPopFunc_t50	Coalescent tree	Constructs a GompertzPopulation_t50 model with optional NA and an l_na indica...

Coalescent(*theta*, *n*, *taxa*, *ages*)

The Kingman coalescent with serially sampled data. (Rodrigo and Felsenstein, 1999)

Parameters:

- Number **theta** - effective population size, possibly scaled to mutations or calendar units.
- Integer **n** - number of taxa.
- Taxa **taxa** - Taxa object, (e.g. Taxa or TimeTree or Object[])
- Double[] **ages** - an array of leaf node ages.

Return type:

- TimeTree

Reference

Rodrigo AG, Felsenstein J. (1999). Coalescent Approaches to HIV Population Genetics, The Evolution of HIV, Chapter 8, edited by Crandall K., Johns Hopkins Univ. Press, Baltimore. <https://books.google.co.nz/books?vid=ISBN0801861519>

Examples

<https://linguaphylo.github.io/tutorials/time-stamped-data/>

Working with Real Data

The Phylogenetic Posterior

Standard application of Bayes theorem gives the posterior:

$$P(T, \mu, \theta | D) = \frac{\Pr(D|T, \mu, \theta)P(T, \mu, \theta)}{\Pr(D)}$$

But most phylogenetic models assume that θ only effects the probability of the data via the tree T , and likewise that μ has no effect on the tree branching process, leading to:

$$P(T, \mu, \theta | D) = \frac{1}{\Pr(D)} \Pr(D|T, \mu)P(T|\theta)P(\mu, \theta)$$

- $P(T|\theta)$ is the "tree prior" parameterized by θ .
- $P(\mu, \theta)$ are the parameter priors.

Questions

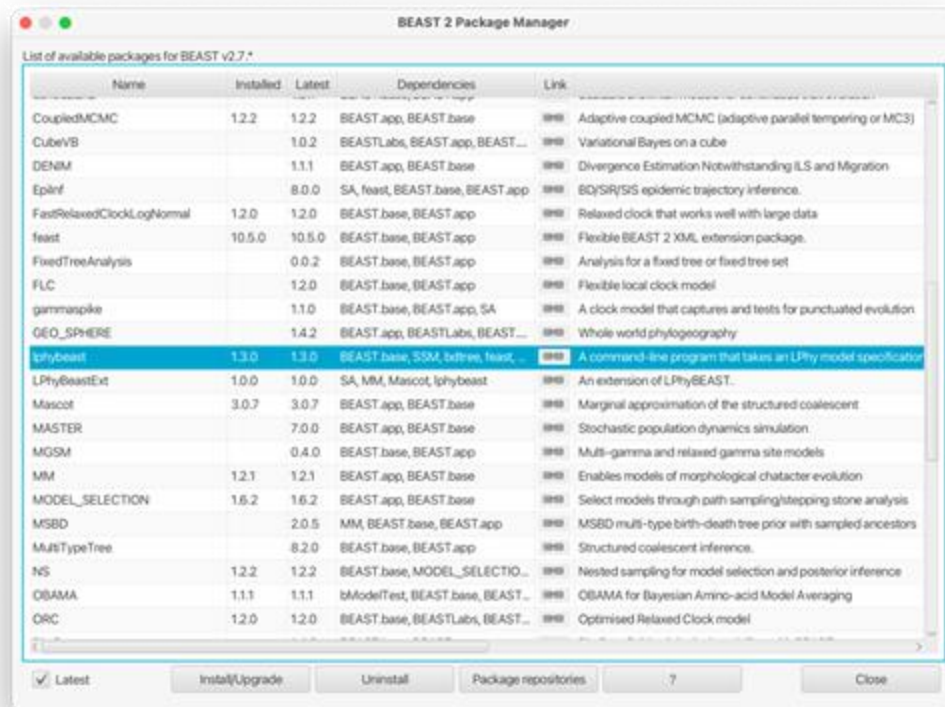
- What is $\Pr(D)$?
- Is the tree prior really a prior? (i.e. does it depend on the data?)

LPhyBEAST

LPhyBEAST is a command-line program that takes a LPhy script file including model specification and data block, and produces a BEAST 2 XML file.

- User manual

<https://linguaphylo.github.io/setup/>

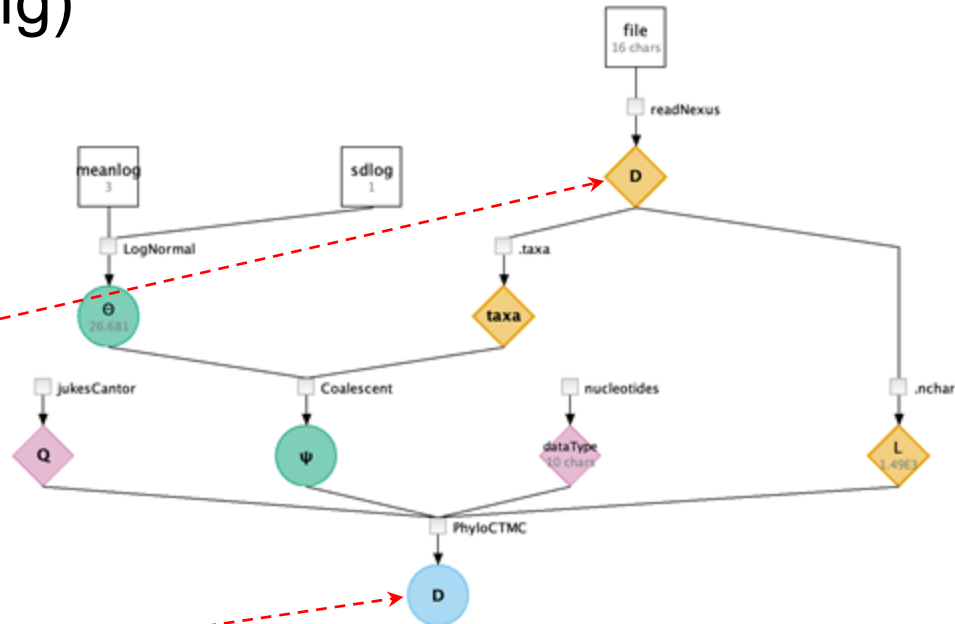


Observation (Data Clamping)

- Alignment
- “Fix” the rate to 1 ?
- “Fix” the tree ?

```
data {  
  D = readNexus(file="data/Dengue4.nex");  
  L = D.nchar();  
  taxa = D.taxa();  
}
```

```
model {  
   $\theta \sim \text{LogNormal}(\text{meanlog}=3.0, \text{sdlog}=1.0);$   
   $\psi \sim \text{Coalescent}(\text{taxa}=\text{taxa}, \text{theta}=\theta);$   
  D ~ PhyloCTMC(L=L, Q=jukesCantor(), dataType=nucleotides(), tree= $\psi$ );  
}
```



LPhyBEAST -ob to “Fix” Parameter

- /Applications/BEAST\ 2.7.8/bin/lphybeast -h

`-ob, --observedParam=<observedParam>[;<observedParam>...]`

Specify one of the many random variables in LPhy as observed using its ID. The specified values will remain fixed during MCMC, including alignments. An ID can be a Unicode symbol, as used in the LPhy script, or its canonical name. Multiple IDs should be separated by ';' without a trailing.

Example: `-ob "A;tree"` or `-ob A`.

If this option is provided with an empty string, all alignments will be sampled during MCMC unless they are defined in the LPhy data block. By default (if this option is not provided), all alignments are assumed to be observed.

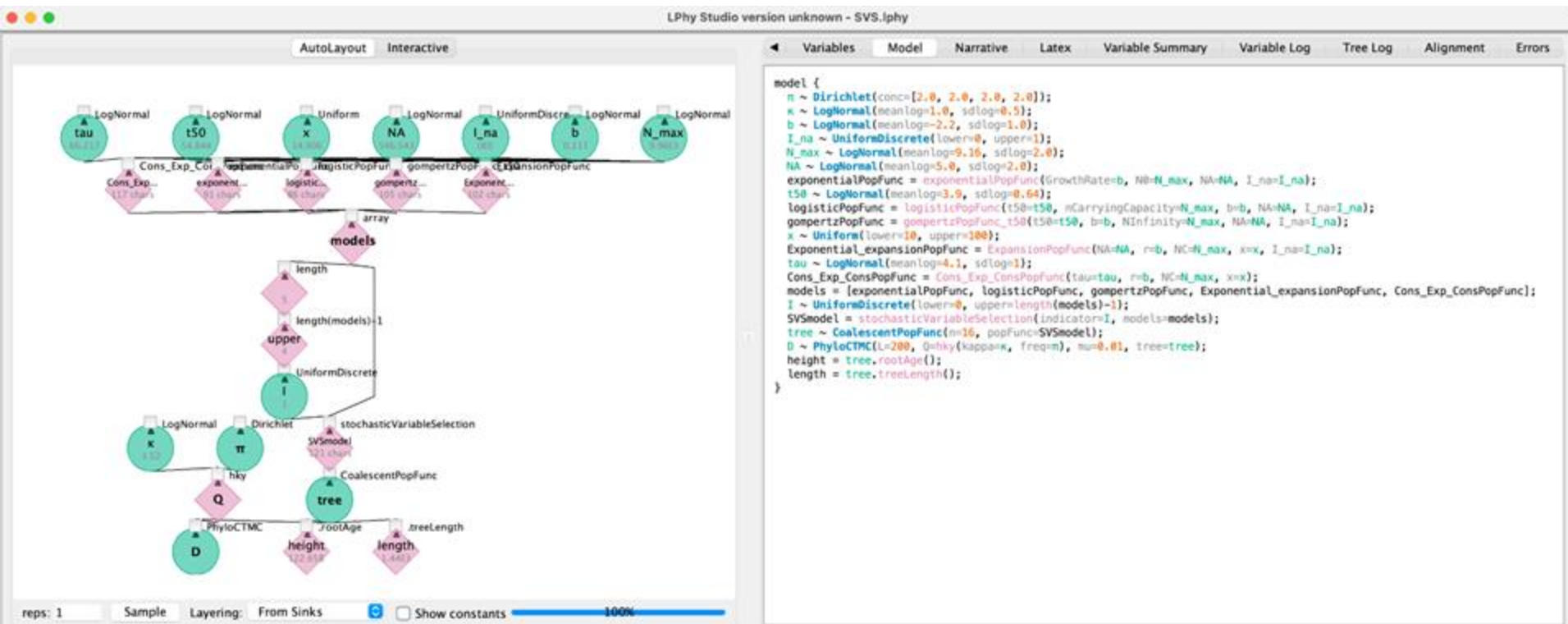
- /Applications/BEAST\ 2.7.8/bin/lphybeast **-ob "psi;D"** jcCoalescent.lphy

Some Experiences & Techniques

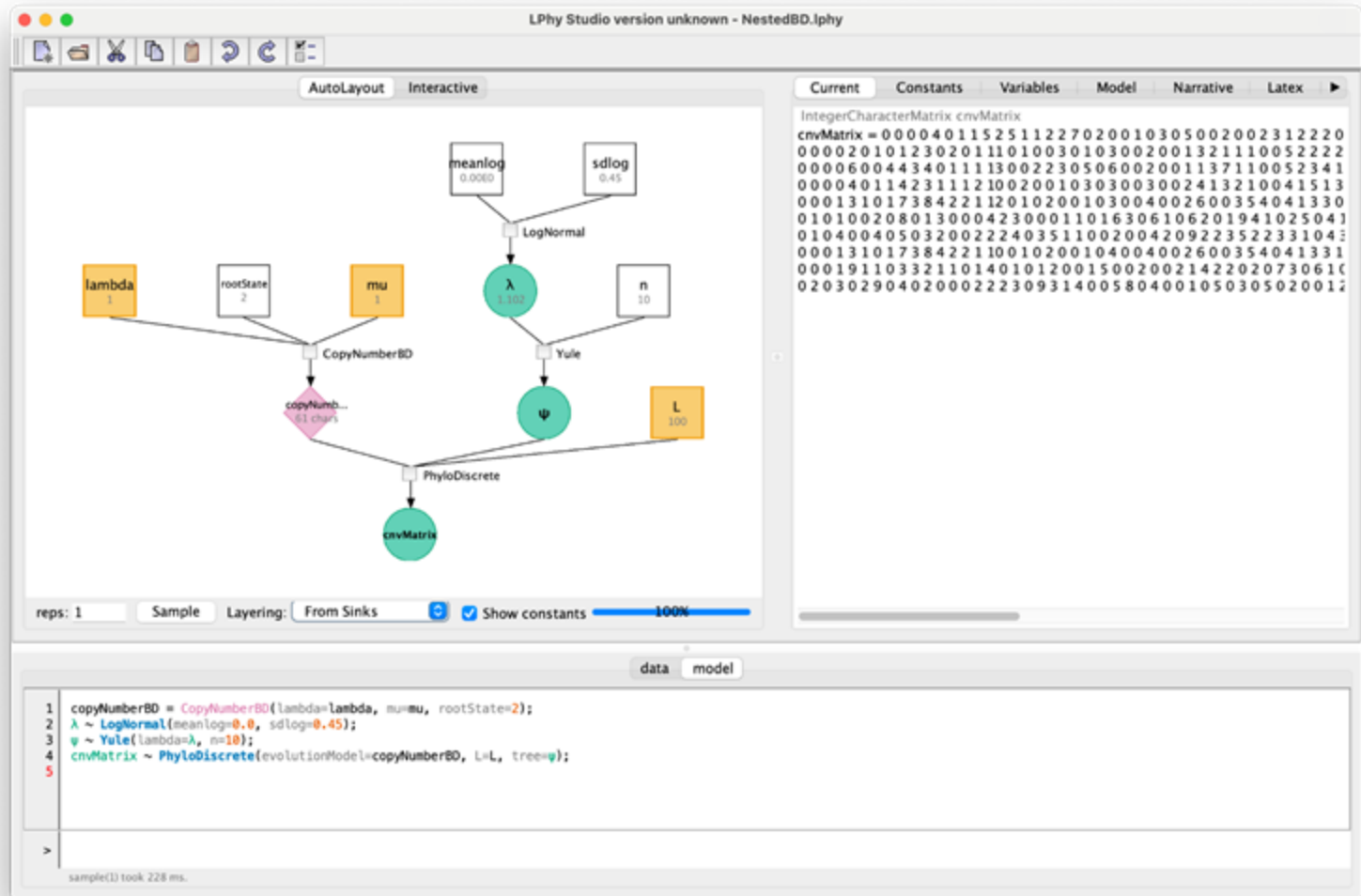
- ❖ Always check your **data** before analysis : garbage in garbage out;
- Every models have their **assumptions** and **limits**;
- All depend on the **parameters of your interests**;
- Never look at result if the MCMC run is not **converged**;
- Model **selection** / averaging
 - bModelTest for substitution models;
 - Nested sampling (NS);
- **Sensitivity** analysis;
- Hypothesis rejection;
- BEAGLE + SSE / multi-CPU / GPU, cloud computing.

Complex Models

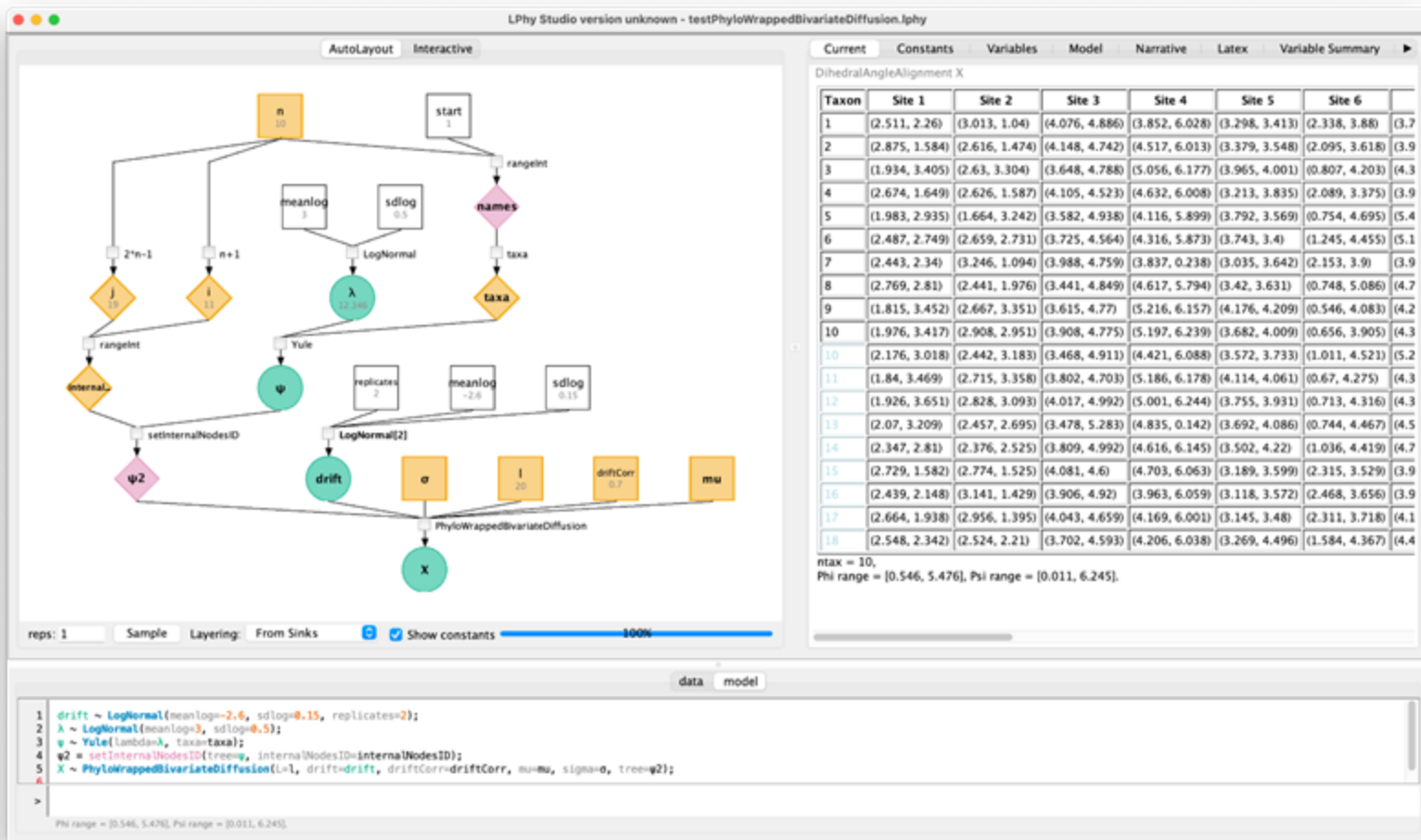
PopFunc : Bayesian Model-Averaging of Parametric Coalescent Models



Nested Birth-death Models for Tumour Phylogenetics



Bayesian Phylogenetic Inference of Protein Structure Evolution using Angular Diffusion Models



LPhy Extensions & Projects

- **ToroidalDiffusion** – Bayesian phylogenetic inference of protein structure evolution using angular diffusion models.
- **PopFunc** – Bayesian model-averaging of parametric coalescent models for phylodynamic inference.
- **ABySS** – Bayesian implementation of time-nonreversible substitution models.
- **Phylonco** – Bayesian inference in cancer evolution, with recent additions:
 - Nested birth-death models for tumour phylogenetics.
 - Joint inference of variants and phylogeny from scDNA-seq read counts.
 - Posterior assessment of single-cell tumour phylogenies.

Future Work

Future Works

1. PhyloSpec : <https://github.com/CODEPhylo/phyloSpec>

It aims to provide a common type system and formal signature definitions for phylogenetic modeling languages.

1. BEAST2Lang : <https://github.com/CompEvol/beast2lang>

A domain-specific language (DSL) for creating statistical phylogenetic models in BEAST2, with a focus on clarity, expressiveness, and statistical modeling principles.

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Next: LPhyBEAST tutorial
“Time-stamped data”